

Behavioral Authentication Under Attack: AI Mouse Bots, Human Impostors, and User-Specific Vulnerability

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Abstract—Continuous behavioral authentication verifies a user throughout a session from signals such as mouse dynamics, but prior evaluations typically test one class of synthetic attacker against one detector and report bypass rates without a reference point. We evaluate per-user mouse-dynamics authentication against three generations of mouse-movement bot—WindMouse, a physics rule; SapiAgent, a trained autoencoder; and DMTG, a diffusion model—each scored against four defender tiers (distance baseline, classical machine learning, tree ensembles, 1D-CNN) on the Balabit Mouse Dynamics Challenge. All attackers are target-blind and zero-query, every configuration is calibrated under identical session-separated splits, and real human-impostor sessions are scored through the same pipeline as a baseline. At the calibrated $\text{FAR} \leq 0.10$ operating point, session bypass is 11.1% for WindMouse, 11.3% for SapiAgent, and 4.2% for DMTG, against 9.4% for real human impostors. Realism and effectiveness are inversely ordered: DMTG is roughly nine times closer to genuine movement on step-level kinematics, yet bypasses least often. Vulnerability is user- and attacker-specific, with largely disjoint vulnerable sets; an attacker free to pick the best tool per target reaches 20.3%. An unsupervised anomaly layer eliminates synthetic bypass but cuts genuine acceptance from 53.7% to 22.6%. In this target-blind setting, bypass is governed by threshold placement rather than trajectory mimicry, and the binding constraint on deployment is false-alarm cost rather than generator sophistication.

Index Terms—behavioral authentication, mouse dynamics, continuous authentication, adversarial robustness, bot detection, NSF REU

I. INTRODUCTION

Continuous authentication detects account misuse after login by monitoring how a user interacts with a system rather than verifying identity only at discrete entry points. Mouse dynamics is an attractive behavioral biometric because it is collected unobtrusively, without extra hardware, and captures user-specific patterns in cursor movement, timing, and navigation [1], [2]. It is already deployed where separating a real user from a bot has stakes: financial account-takeover protection, government access control, and game anti-cheat.

The same machine-learning capability that enables behavioral authentication also creates a new concern. If a model can learn to distinguish legitimate users from impostors, another model may learn to simulate behavior that appears legitimate—an “AI vs. AI” evasion problem [3], [4]. Tan et al. showed that generated mouse sequences degrade remote

authentication in a black-box setting [5], and synthetic trajectory generation has advanced from physics rules [6] through GANs [7] to autoencoders and diffusion models [8], [9].

Two gaps limit what existing evaluations can conclude. First, most studies score a single class of synthetic attacker against a single detector, so a reported bypass rate has no reference point: 10% bypass means something very different if real human impostors bypass 1% of sessions than if they bypass 10%. Second, evaluation hygiene is rarely audited; window-level splits that leak session-specific behavior into training can change the security conclusions, as we show directly. This NSF REU project addresses both gaps with a graded, human-anchored, leakage-controlled evaluation on the Balabit Mouse Dynamics Challenge [10] (Fig. 1). We make four contributions:

- a graded attack benchmark scoring three generations of target-blind mouse bot against four defender tiers under identical, session-separated, leakage-audited calibration;
- a human-impostor baseline scored through the same pipeline, showing that no evaluated bot substantially outperforms a real human impostor;
- evidence that bypass is governed by per-user threshold placement rather than trajectory mimicry, with largely disjoint vulnerable user sets across attackers; and
- a joint security–utility analysis showing that a threshold floor is a targeted fix rather than a general improvement, and that an anomaly layer eliminates synthetic bypass only at an impractical genuine-user cost.

II. RELATED WORK

Mouse-dynamics authentication. Prior work treats continuous mouse-dynamics authentication as per-user verification [1], [2], [11]: Shen et al. [2] report 8.74% false acceptance and 7.69% false rejection, and Siddiqui et al. [11] find a per-user 1D-CNN best among classical and deep models at roughly 86% accuracy on 40 users. Our defenders adopt this formulation across four capacity tiers, but where these studies measure accuracy under normal conditions, we measure robustness when the impostor is *synthetic*.

Synthetic trajectory generation. WindMouse is a gravity-and-wind model for human-like cursor motion [6]; learning-based generators include GAN-based BeCAPTCHA-Mouse [7], the autoencoder SapiAgent [8], and diffusion-based

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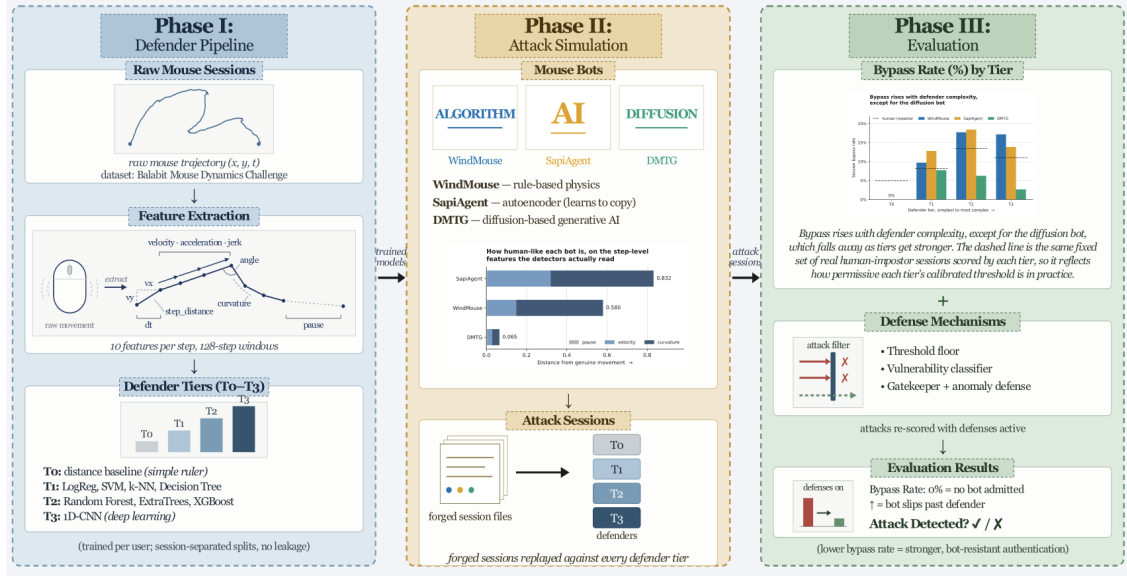


Fig. 1. Evaluation pipeline. Phase I: Balabit sessions are converted to per-event kinematic features, windowed, and used to train four per-user defender tiers (T0–T3) under session-separated, leakage-audited splits. Phase II: three generations of target-blind mouse bot generate forged sessions that are replayed through the identical feature pipeline against every tier. Phase III: session-level bypass is scored at calibrated per-user thresholds, and defenses are evaluated jointly for security and genuine-user utility. The implemented model input is the eight per-event features of Section III-B.

DMTG [9], which reduces bot-detection accuracy by 4.75–9.73% on its CAPTCHA benchmark. These generators target generic human-likeness or bot detection, not impersonation of a specific enrolled user, which is the task we study.

Attacks and explainable defenses. Most relevant to our defense direction are adversarial attacks on behavioral biometrics [5] and the broader adversarial-ML literature on evasion and hardening [4]. Feature attribution identifies which features a defender relies on, and therefore which are most exploitable and most worth hardening; we follow this direction with SHAP [12].

III. METHODOLOGY

A. Problem Formulation and Threat Model

Task. For each enrolled user u , a classifier f_u maps a behavior window x to a score $s(x) = f_u(x) \in [0, 1]$; the system accepts x as genuine when $s(x) \geq \tau_u$, with τ_u a per-user threshold. Training uses windows from u as positives and windows from all other enrolled users, pooled and randomly subsampled to match the positive count, as negatives, giving a balanced 1:1 per-user set with no class weighting. This matches deployment, where the system repeatedly decides whether current activity still comes from the legitimate user.

Attacker knowledge. The attacker is target-blind and zero-query: it has no access to the defender architecture, feature pipeline, threshold τ_u , enrolled-user data, or model feedback. It emits generic human-like cursor motion toward arbitrary targets, and an attack succeeds when a synthetic trajectory scores above τ_u . This is a deliberate lower bound: an attacker this weak that still clears authentication exposes a vulnerability intrinsic to the operating point rather than one manufactured by adaptation; an adaptive attacker with defender feedback

would be expected to do at least as well, but that setting is outside this study.

B. Behavioral Feature Representation

For each event (x_i, y_i, t_i) we compute horizontal and vertical velocity (v_x, v_y) , speed, acceleration, jerk, movement angle, curvature, and a binary pause indicator ($\Delta t > 100$ ms), giving eight features per event in fixed order. Genuine, impostor, and synthetic trajectories pass through identical extraction and clipping. Small inter-event gaps make derivatives unstable, so after replacing non-finite values with zero we apply fixed physical-plausibility caps (not data-driven percentiles): $v_x, v_y \in [-5000, 5000]$ px/s, speed $\in [0, 5000]$ px/s, acceleration $\in [-10^6, 10^6]$ px/s², jerk $\in [-10^8, 10^8]$ px/s³, curvature $\in [-100, 100]$ rad/px; angle is bounded to $[-\pi, \pi]$ by construction. Curvature is the change in movement angle divided by step distance (floored at 10^{-6} px) and is unstable on near-zero-length steps, a point we return to in the realism analysis. Each session is split into non-overlapping windows of 128 events, each a 128×8 authentication sample; the final partial window is discarded. We did not sweep the window length.

C. Per-User Authentication Models

Defender tiers. We evaluate four tiers of increasing capacity, each trained per user: T0, a distance-profile baseline; T1, classical models (logistic regression, SVM, k -NN, decision tree); T2, tree ensembles (Random Forest, ExtraTrees, gradient boosting, XGBoost); and T3, a per-user 1D-CNN. Tabular tiers summarize the per-event features into per-window statistics; the CNN reads the raw 128×8 window. The tiered design tests whether defender

capacity buys robustness, not only clean accuracy. The CNN is Conv1D(64)→BN→MaxPool→Conv1D(128)→BN→MaxPool→GAP→Dense(128)→Dropout(0.5)→Dense(1, sigmoid), kernel size 3, stride 1, valid padding, ReLU, pool size 2, trained with binary cross-entropy and Adam (learning rate 10^{-3} , batch 64, 20 epochs, no early stopping), identical across users and seeds $\{0, 1, 2\}$.

Training protocol. Data are split at the session level, never the window level, so windows from one session cannot span train and test. The train/test division follows the Balabit challenge’s predefined split: genuine `training_files` for training and the labeled `test_files`, containing genuine and human-impostor sessions, for evaluation. Within the training pool we hold out 25% as a stratified calibration set and fit on the remaining 75% (for the CNN, a further 20% of the fit set is validation). A `StandardScaler` is fit per user on training windows only and applied unchanged to calibration, test, and synthetic windows.

Decision rule. A fixed 0.5 threshold suits few users, since score distributions shift across enrolled users. We compare four policies: **default** ($\tau_u = 0.5$); **equal-error-rate**; **low-FAR** (lowest calibration threshold holding calibration $\text{FAR} \leq 0.10$); and **floored low-FAR** (low-FAR clamped to a minimum of 0.05, preventing a degenerate near-zero threshold when a user’s calibration scores collapse, Section IV-B). All calibrated thresholds are computed on the calibration set only and applied unchanged to held-out test and synthetic sessions. Authentication-quality metrics are reported at the window level; for the attack we define a session-level decision (a session is accepted when at least half its windows score at or above τ_u), and session-level bypass at the calibrated low-FAR threshold is our headline security metric.

D. Target-Blind Synthetic Attacks

Three attacker generations. *WindMouse* [6] produces cursor motion from gravity toward a target plus random wind forces and velocity limits, run at its published defaults ($G=9$, $W=3$, $M=15$, $D=12$). *SapiAgent* [8] is an autoencoder trained to generate human-like trajectories from real movement data. *DMTG* [9] is an entropy-controlled diffusion model, run without its style-transfer module so generation remains target-blind. None of the three learns from the enrolled users or optimizes against the authenticator. *SapiAgent* and *DMTG* follow their published formulations; the diffusion attacker was trained locally in PyTorch, and the generation settings for both learned attackers accompany the project code.

Generation and sampling. Generated trajectories are rendered into the Balabit raw event format and pass through the same extraction, clipping, windowing, and normalization as genuine and impostor data. We generate 20 sessions per seed over seeds $\{0, 1, 2\}$ (60 per attacker), each of 12 movement segments whose start and target points are sampled uniformly over a 1920×1080 screen, so no enrolled-user geometry enters generation; a single target-blind set per attacker is scored against every user. Inter-event timings are sampled from

a pool of gaps concatenated across all 65 Balabit training sessions ($0 < \Delta t \leq 1.0$ s), with inter-segment pauses from Uniform(0.05, 0.6) s; the pooled timing leaks no per-user information. Bypass is the fraction of synthetic sessions accepted as genuine.

Sanity baseline. On the CNN at the low-FAR threshold (window level, averaged over ten users and three seeds), genuine TAR is 0.751, human-impostor FAR is 0.207, and WindMouse bypass is 0.249—at or above the human-impostor rate. The blind synthetic attacker clears authentication at least as often as a real different user attempting impersonation, so it is a meaningful lower-bound attacker, not a strawman.

E. Evaluation Protocol

For normal operation we report TAR, FAR, FRR, AUC, and EER; for the attack, window- and session-level bypass rate, mean attack score, attack-score headroom (mean attack score minus threshold), and a feature-level realism comparison. All metrics are averaged over the ten enrolled users, and because bypass concentrates in a few users we also describe its distribution rather than relying on the mean alone. Defender training and attack generation both use seeds $\{0, 1, 2\}$; reported numbers are three-seed averages, and tier-averaged numbers weight the four tiers equally. The realism comparison reports per-feature session-level means across the genuine and attacker populations; because curvature is unstable on near-zero-length steps we report it restricted to steps with displacement ≥ 2 px, and we do not reduce realism to a single distributional distance, since a scalar built on unfiltered curvature would spuriously rank the slowest attacker as most human. Attackers were built and defenders scored in Colab, except the diffusion attacker, trained locally in PyTorch on an RTX 5070.

IV. RESULTS

We organize the results around four findings. All numbers are over ten enrolled users and three seeds, and each traces to a released result artifact.

A. Session-separated evaluation changes the security conclusions

We corrected two preprocessing and splitting errors present in earlier runs. First, records sharing a timestamp were previously assigned velocity, acceleration, and jerk under a $\Delta t = 10^{-6}$ substitution, inflating these features to their clipping bounds; duplicate-timestamp rates reach 0.96 for some users (e.g., user9, user20). We now collapse duplicate timestamps before feature extraction. Second, and more consequential, windows from a single session previously appeared in more than one of the fit, calibration, and test splits. We separate splits at the session level and verify zero session overlap across the fit/calibration, fit/test, calibration/test, and CNN train/validation partitions for all ten users and three seeds. Under the corrected protocol the strongest clean tier is the T2 tree ensemble (AUC 0.865, EER 0.197), ahead of the T3 1D-CNN (AUC 0.812, EER 0.245; Table I).

TABLE I
CLEAN DEFENDER COMPARISON AT THE DEFAULT THRESHOLD OF 0.5
(MEAN OVER USERS AND SEEDS). AUC AND EER ARE
THRESHOLD-INDEPENDENT.

Tier / Model	TAR	FAR	AUC	EER
T0 (Distance profile)	0.830	0.612	0.673	0.362
T1 (LogReg/SVM/kNN/tree)	0.664	0.174	0.799	0.243
T2 (Tree ensembles)	0.627	0.099	0.865	0.197
T3 (1D-CNN)	0.617	0.174	0.812	0.245

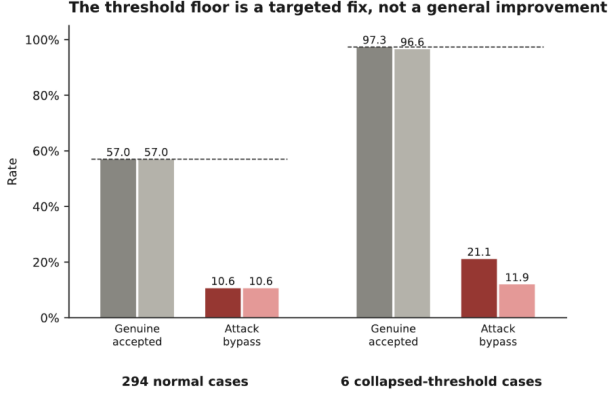


Fig. 2. Genuine acceptance and mean attack bypass before (dark) and after (pale) a 0.05 threshold floor, split by whether the calibrated threshold collapsed below 0.05. Case-level means over 300 model-user-seed combinations; 5 of 6 collapsed cases are T2, one is T3.

B. Strong clean performance does not guarantee robustness

The clearest case is a T2 tree ensemble for user7: it attains a near-perfect clean AUC of 0.9998, yet its calibrated low-FAR threshold collapses below 0.05 on two of three seeds (to 0.047 for gradient boosting and 0.003 for XGBoost), at which point SapiAgent bypasses 40–70% of sessions while genuine acceptance stays above 0.99. The remaining seed does not collapse (threshold ≈ 0.26 , zero bypass): the failure is reproducible but seed-sensitive. A model can thus separate genuine from impostor traffic almost perfectly and still admit a target-blind synthetic attacker, because bypass is governed by where the threshold lands, not by clean separability.

Enforcing a floor on the calibrated threshold is a targeted remedy, not a general defense (Fig. 2). Across all user, model, and seed configurations only 1.7% change at a floor of 0.05 and 7.3% at 0.20, at negligible genuine-TAR cost. The floor closes the user7 collapse (SapiAgent bypass falls to 0% by a floor of 0.20, TAR 0.996 $\rightarrow$ 0.971), but at the highest floor at least one configuration still bypasses fully, e.g., a user21 linear model (SapiAgent 100% at TAR 0.70) and a user23 tree ensemble (WindMouse 100% at TAR 0.60).

C. Attack vulnerability is user- and attack-specific

No single attacker dominates. Averaging over the four defender tiers at the calibrated $\text{FAR} \leq 0.10$ operating point, mean session bypass is 11.1% for WindMouse, 11.3% for SapiAgent, and 4.2% for DMTG, against a 9.4% real-human-

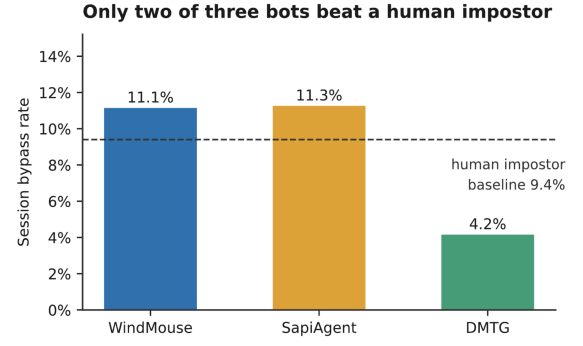


Fig. 3. Session bypass by attacker, averaged over the four defender tiers, against the real human-impostor baseline of 9.4% (dashed); calibration $\text{FAR} \leq 0.10$, no floor. Only two of three bots exceed the human baseline.

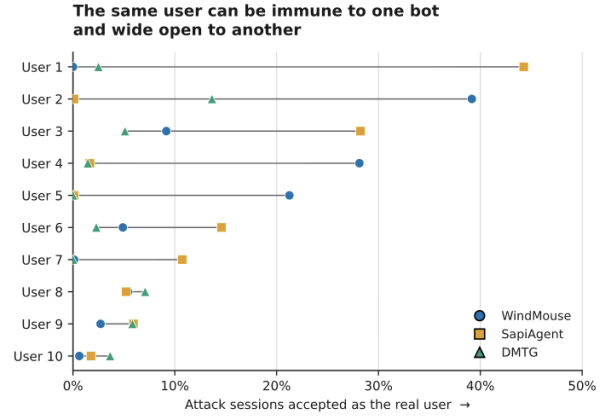


Fig. 4. Per-user session bypass by attacker (tier-averaged), users sorted by worst-case bypass. User 1–10 correspond to Balabit users 21, 15, 29, 23, 9, 7, 35, 20, 16, and 12 in Table II. The most effective bot differs from user to user, so vulnerable sets are largely disjoint.

impostor baseline (Fig. 3). At the user level ($n=10$, the unit for our statistics), WindMouse and SapiAgent are indistinguishable (permutation $p=0.783$); WindMouse significantly exceeds DMTG ($p=0.036$), whereas SapiAgent versus DMTG does not reach significance ($p=0.128$), so we claim only the single WindMouse-over-DMTG result rather than a general ranking.

The per-user breakdown (Table II) shows the specificity directly: user15, user23, and user9 fall to WindMouse at near-zero SapiAgent bypass (≤ 0.02), whereas user21, user7, user35, and user16 fall to SapiAgent at near-zero WindMouse bypass; user29 is vulnerable to both, and DMTG reaches non-negligible bypass only for user15 (0.136). Fig. 4 plots the same data sorted by worst-case bypass: in Table II the gap between the most and least effective bot exceeds 0.10 for seven of ten users, and the identity of the most effective bot alternates down the ranking. An attacker free to choose the best tool per target succeeds 20.3% of the time, nearly double the best single attacker, although this assumes a selection step we did not implement.

Attacker realism does not explain effectiveness (Table III). Inter-event timing does not separate the populations—all three

TABLE II
PER-USER SESSION BYPASS BY ATTACKER, AVERAGED OVER THE FOUR DEFENDER TIERS AND SEEDS AT THE CALIBRATED FAR ≤ 0.10 OPERATING POINT. THE SAME USER CAN BE IMMUNE TO ONE BOT AND WIDE OPEN TO ANOTHER; COLUMN MEANS EQUAL THE REPORTED AGGREGATE.

User	WindMouse	SapiAgent	DMTG
user15	0.392	0.001	0.136
user23	0.281	0.017	0.015
user9	0.212	0.001	0.000
user29	0.092	0.282	0.051
user20	0.054	0.052	0.071
user7	0.049	0.146	0.023
user16	0.027	0.059	0.058
user12	0.006	0.018	0.036
user35	0.001	0.107	0.000
user21	0.000	0.443	0.025
Mean	0.111	0.113	0.042

TABLE III
TRAJECTORY REALISM VERSUS GENUINE BALABIT MOVEMENT, SAME EXTRACTOR AND DUPLICATE-TIMESTAMP COLLAPSE AS THE DEFENDER PIPELINE. PAUSE: FRACTION OF STEPS WITH $\Delta t > 0.1$ s; CURVATURE: MEAN $|\Delta\theta/d|$ ON DISPLACEMENT ≥ 2 px. GENUINE $n=65$ SESSIONS; EACH ATTACKER $n=60$. LAST ROW IS THE MEAN ABSOLUTE LOG-RATIO TO GENUINE ACROSS THE THREE TABULATED METRICS (LOWER IS MORE HUMAN-LIKE).

Metric	Genuine	WindMouse	SapiAgent	DMTG
Pause rate	0.844	0.853	0.854	0.854
Velocity (px/s)	570	369	221	528
Curvature (≥ 2 px)	0.074	0.020	0.345	0.083
Mean $ \log\text{-ratio} $	—	0.585	0.833	0.068

attackers share the empirical timing sampler and reach pause rates of 0.853–0.854 against a genuine 0.844—but geometry does: on displacement of at least two pixels, SapiAgent’s curvature is $4.6\times$ genuine (0.345 vs. 0.074), whereas DMTG sits at $1.11\times$ (0.083). Across all three metrics DMTG is by far the closest attacker to genuine movement (mean absolute log-ratio to genuine 0.068, against 0.585 for WindMouse and 0.833 for SapiAgent, computed from the tabulated values), yet it is the least effective attacker in aggregate. Realism and bypass success are not merely uncorrelated but inversely ordered here, the reverse of what a mimicry account predicts.

D. Defense effectiveness must include genuine-user utility

Adding an unsupervised Isolation Forest anomaly layer behind the per-user gate (T2 Random Forest) drives synthetic bypass to zero at a genuine-user cost that makes it impractical as deployed (Fig. 5): final synthetic-attack acceptance falls from 0.6% to 0.0%, but the layer flags 58.4% of the genuine sessions the gate had accepted, and genuine acceptance drops from 53.7% to 22.6%. The layered scheme is therefore a security–usability tradeoff rather than a successful final defense. We report both quantities for every defense so that robustness gains are never credited without their utility cost.

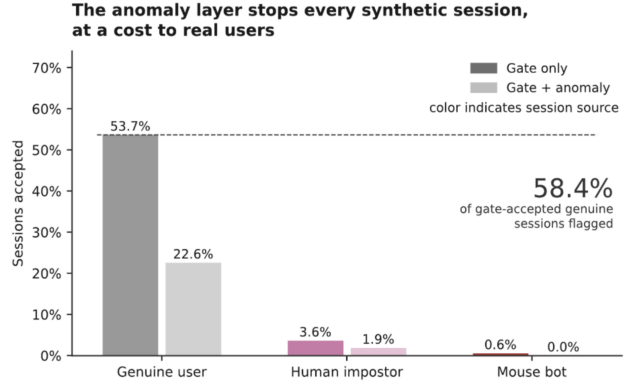


Fig. 5. Session acceptance at the gatekeeper (T2 Random Forest) and after the Isolation Forest anomaly layer, by session source; per-user means. The anomaly layer stops every synthetic session at the cost of flagging 58.4% of gate-accepted genuine sessions.

E. Human-impostor baseline and exploratory analyses

Real impostor sessions from other enrolled users (the A0 baseline) give the synthetic attackers a reference point. The tier-averaged human-impostor session FAR used in Fig. 3 is 9.4%; the dedicated A0 session artifact reports a session FAR of 10.2% at a genuine session TAR of 63.5%. We use 9.4% wherever it is compared with bypass rates because it is computed with the same tier averaging, and the qualitative result is unchanged at 10.2%: WindMouse and SapiAgent remain above the human baseline and DMTG below it. In either view the defenders are calibrated to a realistic human-impostor error rate.

Three further analyses are exploratory and reported as hypotheses. **Mechanism:** a blind-spot margin on whole-session features correlates only weakly with bypass and does not reach significance at our scale (Spearman $\rho=0.294$, $n=30$ user-by-attacker cells). **Vulnerability classifier:** a leave-one-user-out classifier predicting which users are vulnerable reaches balanced accuracy 0.617 at the loosest cutoff but falls to 0.53–0.60 at moderate cutoffs (chance 0.50); the signal is weak and not robust to the definition of vulnerable. **Feature attribution:** a per-user SHAP case study identifies curvature-based features as the top attribution for the WindMouse- and SapiAgent-dominant users, stable across seeds; user21’s clean-side defender was weak (high FAR), so part of its bypass reflects general weakness rather than an attacker-specific signature.

V. DISCUSSION

Bypass is exploitation, not mimicry. WindMouse bypasses three users despite being distinguishable from genuine movement in aggregate; aggregate separability and per-user acceptance are different properties. A threshold calibrated against human impostors leaves an acceptance region that a non-mimicking synthetic input can fall into.

Good defenders can be more exposed. The most surprising case is user9, a flawless authenticator against humans (AUC 1.0, EER 0.0) nonetheless bypassed on 93% of attack sessions by one configuration: clean genuine/impostor separation lets

the $FAR \leq 0.10$ calibration drop the threshold almost to zero, enlarging the acceptance region and admitting WindMouse. With one such case among ten users this is an existence proof, not a general law.

Defense directions. Because WindMouse is separable in aggregate, a defender that rejects out-of-distribution inputs, trains against synthetic negatives, or hardens the most exploitable features (speed and curvature, where WindMouse differs most) should recover robustness, motivating an attribution-guided feature-hardening direction [12].

Deployment reading. Two design choices in the evaluation carry over to practice. First, the session-level rule (accept when at least half of a session’s windows pass) is the natural unit for continuous authentication, where an impostor must sustain acceptance over time rather than pass a single check; window-level bypass rates (e.g., 0.249 for WindMouse on the CNN) overstate what a bot achieves at the session level. Second, because vulnerability is user-specific, a deployment cannot certify robustness from an aggregate bypass rate: per-user calibration audits, in particular flagging users whose calibrated threshold falls far below the population, are a cheaper first line than a global anomaly layer, and the threshold-floor result shows they would target exactly the cases that fail.

Reproducibility. Every reported number traces to a project result artifact (clean defender results, attack results, threshold-floor sensitivity, realism metrics, and layered-defense session results), the defender and attack seeds are fixed at $\{0, 1, 2\}$, and the session-split audit recording zero overlap is kept alongside them. Because bypass is heavy-tailed across users, we recommend that follow-up work report per-user distributions alongside means, as in Table II; a mean of 11% conceals that most users are near 0% and a few are near 40%.

Limitations. Results are on ten users of one dataset, and with $n=10$ the headroom correlation is directional: it should be read as “the attacker’s score distribution relative to the operating point governs bypass,” not as a discovered constant. Our attackers sample per-event timing independently of step size, breaking the speed–displacement coupling of real movement (Fitts’s law), which is why WindMouse comes out slow and pause-heavy; conditioning timing on displacement is future work. All thresholds were calibrated for security alone ($FAR \leq 0.10$); calibrating toward usability is untested.

VI. CONCLUSION

Across ten enrolled users, four defender tiers, and three generations of mouse-movement bot, more sophisticated generation did not, without target conditioning, produce more effective attacks: the diffusion attacker is by far the closest to genuine movement on the step-level features the defenders read, yet it bypassed least often and never reached the real human-impostor rate. Bypass tracked where each user’s decision threshold landed rather than how human-like the trajectory was, and vulnerability was specific to particular user-and-attacker pairs. Read as a deployment question the result is two-sided: against synthetic impersonation, per-user behavioral authentication held up better than the premise assumed, but the

layered defense that eliminated synthetic bypass also flagged 58.4% of the genuine sessions it had admitted, reducing genuine acceptance to 22.6%. For the attackers evaluated here, the binding constraint is false-alarm cost, not generator sophistication. These conclusions are bounded by ten users of a single dataset and by target-blind, zero-query attackers; an adaptive attacker that observes defender feedback would be expected to reach higher rates and is not evaluated here.

Three directions follow: normalizing threshold calibration across model families, since a raw floor is not comparable across linear, tree, and convolutional score distributions; calibrating the anomaly layer on larger per-user session samples; and extending the diffusion attacker with the omitted style-transfer module, preserving target-blindness, to test whether target conditioning separates our result from the generator’s published performance.

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